SUPPLEMENTAL MATERIAL: Forest Plots of Secondary Outcomes

1) Postoperative NAHS

Figure S1 demonstrates the difference in retroversion, anteversion and abnormal groups in postoperative NAHS when compared with normal group. Patients with femoral retroversion had significant difference at postoperative NAHS scores (2 studies, MD, 5.96 [95% CI, 1.66 to 10.26], *P*= .007), with no significant heterogeneity (I^2= 0%, *P*= .051). Patients with femoral anteversion had no significant difference (2 studies, MD, -4.39 [95% CI, -19.27 to 10.49]; *P*= .56), with significant heterogeneity (I^2= 70%, *P*= .07). Besides, overall effect of abnormal FV group showed significant difference (3 studies, MD, 2.58 [95% CI, -3.11 to 8.28]; *P*= .37), with significant heterogeneity (I^2= 56%, *P*= .08).

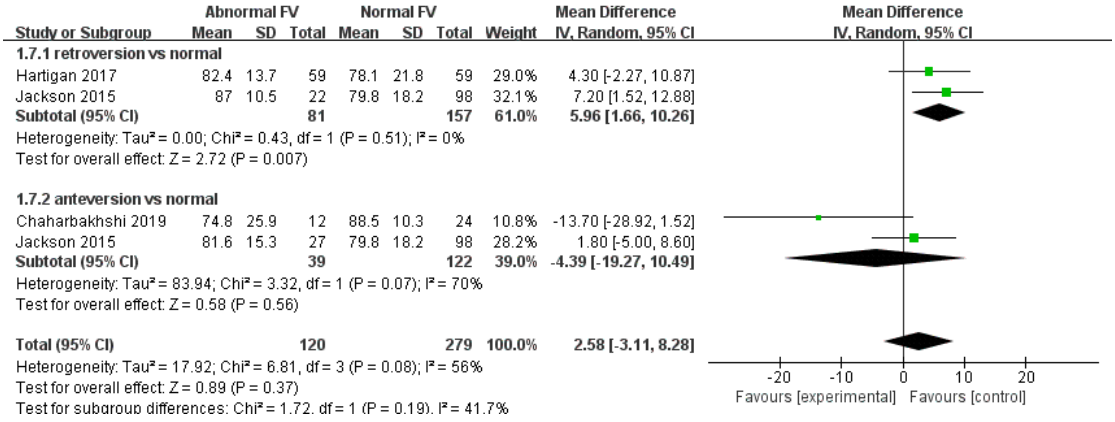


Figure S1. Forest plot of postoperative NAHS.

2) Δ_{NAHS}

Figure S2 demonstrates the difference in retroversion, anteversion and abnormal groups in Δ_{NAHS} when compared with normal group. Patients with femoral retroversion had no significant difference at Δ_{NAHS} scores (2 studies, MD, 3.79 [95% CI, -1.66 to 9.24]; *P*= .17), with no significant heterogeneity (I^2= 0%, *P*= .35). Patients with femoral anteversion had no significant difference (2 studies, MD, -2.35[95% CI, -8.23 to 3.53]; *P*= .43), with no significant heterogeneity (I^2= 0%, *P*= .62). Besides, overall effect of abnormal FV group showed no

significant difference (3 studies, MD, 0.96 [95% CI, -3.33 to 5.26]; *P*= .66), with significant heterogeneity (I^2= 11%, *P*= .34).

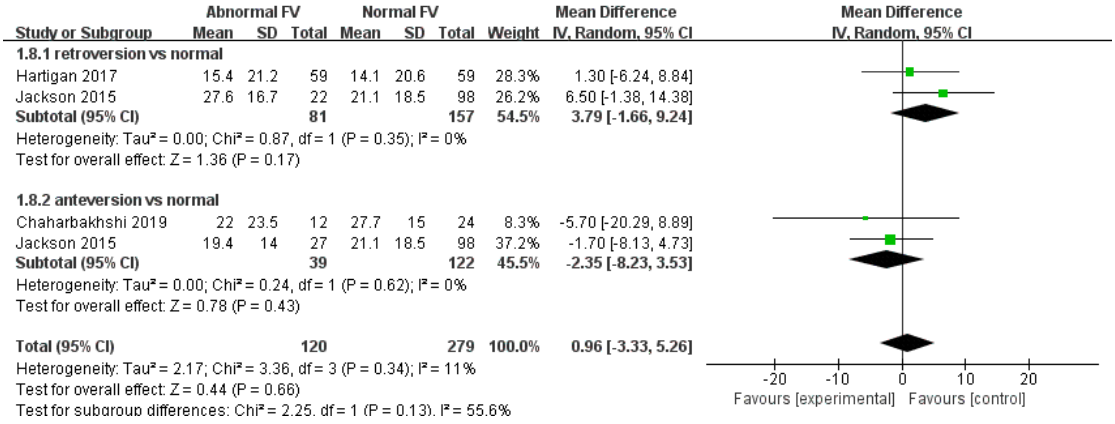


Figure S2. Forest plot of Δ_{NAHS}.

3) Postoperative HOS-SSS

Figure S3 demonstrates the difference in retroversion, anteversion and abnormal groups in postoperative HOS-SSS when compared with normal group. Patients with femoral retroversion had significant difference at postoperative HOS-SSS scores (2 studies, MD, 7.32 [95% CI, 0.19 to 14.44]; *P*= .04), with no significant heterogeneity (I^2= 0%, *P*= .46). Patients with femoral anteversion had no significant difference (2 studies, MD, -6.49 [95% CI, -29.12 to 16.13]; *P*= .57), with significant heterogeneity (I^2= 70%, *P*= .07). Besides, overall effect of abnormal FV group showed no significant difference (3 studies, MD, 2.77 [95% CI, -6.04 to 11.58]; *P*= .54), with no significant heterogeneity (I^2= 49%, *P*= .12).

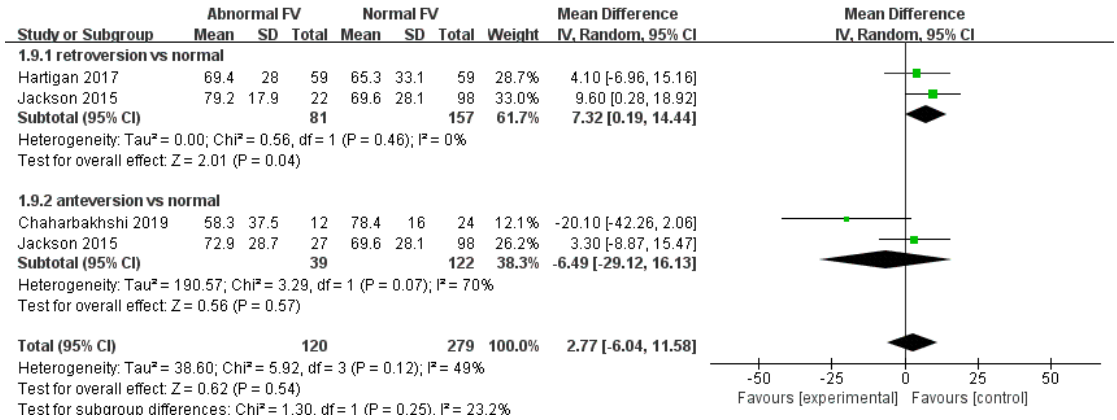


Figure S3. Forest plot of postoperative HOS-SSS.

4) $\Delta_{HOS\text{-}SSS}$

Figure S4 demonstrates the difference in retroversion, anteversion and abnormal groups in $\Delta_{HOS\text{-}SSS}$ when compared with normal group. Patients with femoral retroversion had significant difference at $\Delta_{HOS\text{-}SSS}$ scores (3 studies, MD, -2.61 [95% CI, -15.72 to 10.50]; *P*= .70), with significant heterogeneity (I^2= 78%, *P*= .01). Patients with femoral anteversion had no significant difference (2 studies, MD, -2.44 [95% CI, -12.25 to 7.37]; *P*= .63), with no significant heterogeneity (I^2= 0%, *P*= .48). Besides, overall effect of abnormal FV group showed no significant difference (4 studies, MD, -3.01 [95% CI, -11.34 to 5.32]; *P*= .48), with significant heterogeneity (I^2= 59%, *P*= .05).

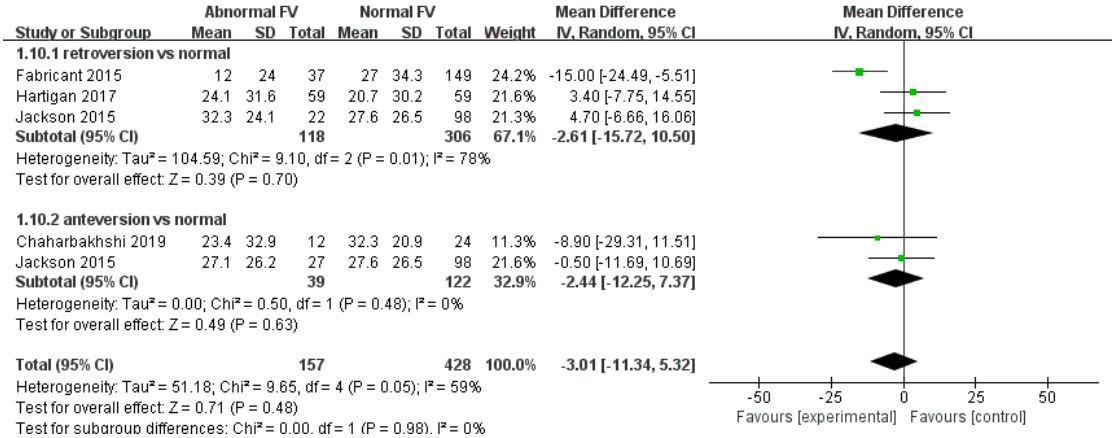


Figure S4. Forest plot of ΔHOS-SSS.

5) Postoperative VAS

Figure S5 demonstrates the difference in retroversion, anteversion and abnormal groups in postoperative VAS when compared with normal group. Patients with femoral retroversion had no significant difference at postoperative VAS (2 studies, MD, -3.82 [95% CI, -9.92 to 2.28]; *P*= .69), with no significant heterogeneity (I^2= 0%, *P*= .44). Patients with femoral anteversion had no significant difference (2 studies, MD, -0.09 [95% CI, -1.29 to 1.47]; *P*= .90), with no significant heterogeneity (I^2= 41%, *P*= .19). Besides, overall effect of abnormal FV group showed no significant difference (3 studies, MD, -0.12 [95% CI, -0.64 to 0.40]; *P*= .66), with no significant heterogeneity (I^2= 0%, *P*= .52).

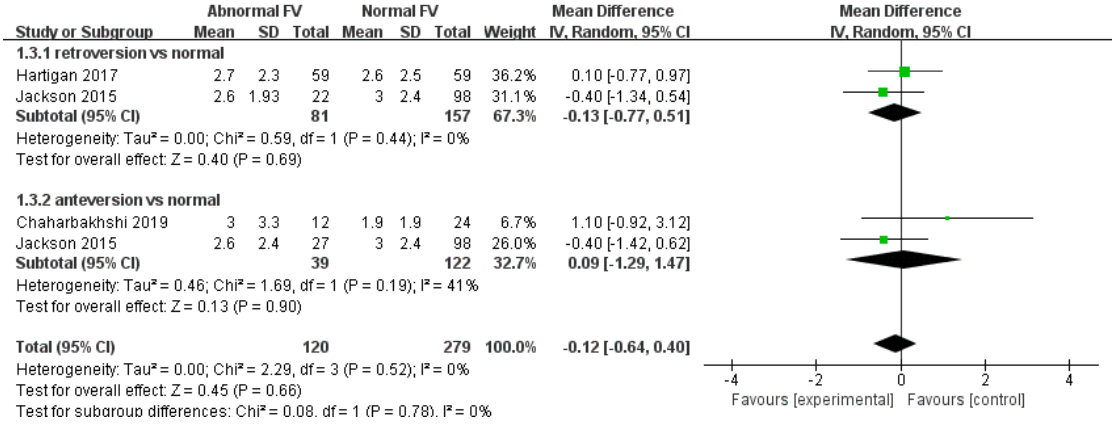


Figure S5. Forest plot of postoperative VAS.

6) Δ_{VAS}

Figure S6 demonstrates the difference in retroversion, anteversion and abnormal groups in Δ_{VAS} when compared with normal group. Patients with femoral retroversion had no significant difference at Δ_{VAS} (2 studies, MD, -0.13 [95% CI, -0.87 to 0.60]; *P*= .72), with no significant heterogeneity (I^2= 0%, *P*= .43). Patients with femoral anteversion had no significant difference (2 studies, MD, -0.19 [95% CI, -2.11 to 1.74]; *P*= .85), with significant heterogeneity (I^2= 71%, *P*= .07). Besides, overall effect of abnormal FV group showed no significant difference (3 studies, MD, -0.27 [95% CI, -1.00 to 0.45]; *P*= .46), with no significant heterogeneity (I^2= 38%, *P*= .19).

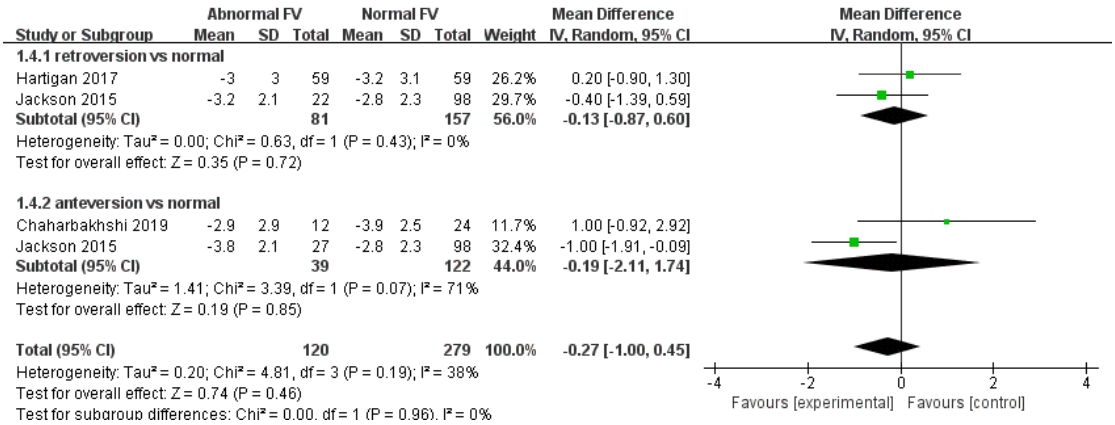


Figure S6. Forest plot of Δ_{VAS}.

7) Patient satisfaction

Figure S7 demonstrates the difference in retroversion, anteversion and abnormal groups in patient satisfaction when compared with normal group. Patients with femoral retroversion had no significant difference at patient satisfaction (1 study, MD, -0.20 [95% CI, -1.49 to 1.09]; *P*= .76), with no applicable heterogeneity. Patients with femoral anteversion had no significant difference (2 studies, MD, -0.11 [95% CI, -1.95 to 1.73]; *P*= .91), with significant heterogeneity (I^2= 72%, *P*= .06). Besides, overall effect of abnormal FV group showed no significant difference (3 studies, MD, -0.05 [95% CI, -1.08 to 0.99]; *P*= .24), with no significant heterogeneity (I^2= 0%, *P*= .82).

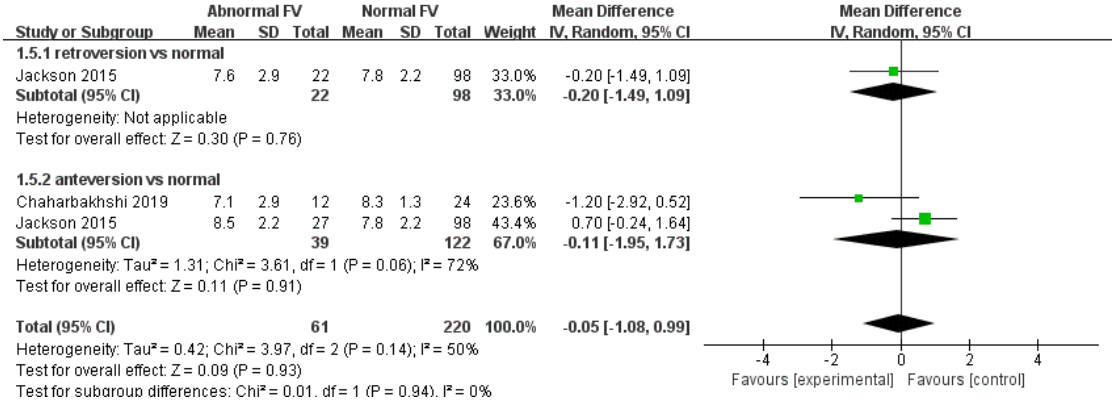


Figure S7. Forest plot of patient satisfaction.

8) Failure rate

Figure S8 demonstrates the difference in retroversion, anteversion and abnormal groups at failure rate when compared with normal group. Patients with femoral retroversion had no significant difference at failure rate (2 studies, MD, 0.56 [95% CI, 0.18 to 1.77]; *P*= .32), with no significant heterogeneity (I^2= 0%, *P*= .91). Patients with femoral anteversion had no significant difference (2 studies, MD, 0.70 [95% CI, 0.25 to 1.91]; *P*= .48), with no significant heterogeneity (I^2= 0%, *P*= .36). Besides, overall effect of abnormal FV group showed no significant difference (3 studies, MD, 0.63 [95% CI, 0.30 to 1.35]; *P*= .24), with no significant heterogeneity (I^2= 0%, *P*= .82).

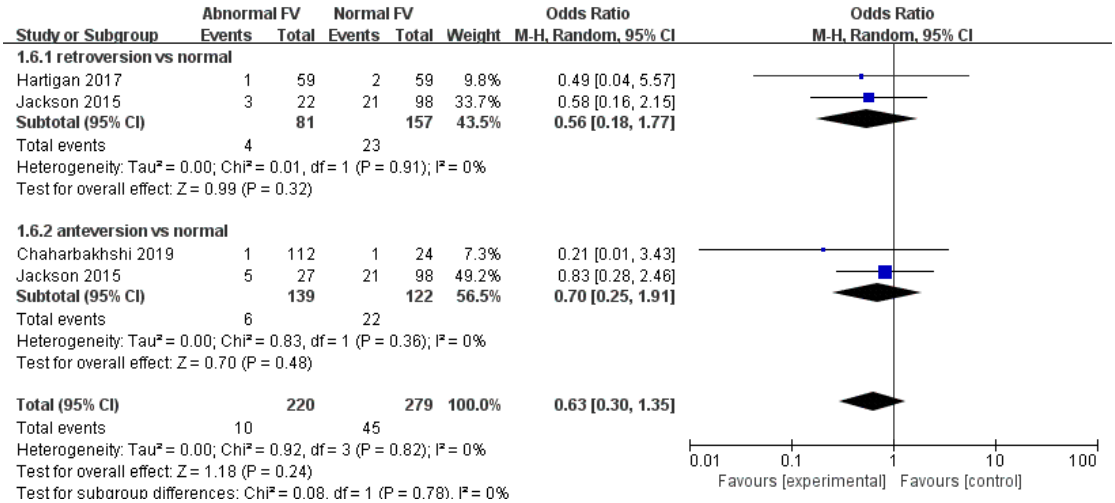


Figure S8. Forest plot of failure rate.